Revision Game Cards — 1.4 Data Types, Data Structures & Boolean Algebra

Print and cut along the borders. ✂️ Loop cards: each player reads the bottom "Who has…?"; whoever holds the matching "I have…" reads it out, then their own clue — the chain visits every card and loops back to the start.

Loop cards (dominoes) — cut out & deal

1 / 15

I HAVE

0xD6

WHO HAS…

the denary value of binary 1010??

2 / 15

I HAVE

10

WHO HAS…

the result of 0 + 0 in binary??

3 / 15

I HAVE

0

WHO HAS…

the number of bits in a nibble??

4 / 15

I HAVE

4

WHO HAS…

the denary value of hex F??

5 / 15

I HAVE

15

WHO HAS…

the denary value of binary 00100000??

6 / 15

I HAVE

32

WHO HAS…

the hex value of binary 1111??

7 / 15

I HAVE

F

WHO HAS...

the data structure that is LIFO??

8 / 15

I HAVE

Stack

WHO HAS...

the data structure that is FIFO??

9 / 15

I HAVE

Queue

WHO HAS...

the 8-bit two's complement of denary -1??

10 / 15

I HAVE

11111111

WHO HAS...

the denary value of binary 00001000??

11 / 15

I HAVE

8

WHO HAS...

the effect of a logical left shift by 1, as a multiplication??

12 / 15

I HAVE

Multiply by 2

WHO HAS...

the law that states NOT(A AND B) = NOT A OR NOT B??

13 / 15

I HAVE

De Morgan's

WHO HAS...

the floating-point part that sets the precision??

14 / 15

I HAVE

Mantissa

WHO HAS...

the value of 0xD6 (214) minus 200, in denary??

I HAVE

14

WHO HAS...

the hex value of denary 214??

Taboo cards — describe the term without the banned words

Two's complement

Don't say:
- negative
- MSB
- invert
- add one

Normalisation

Don't say:
- mantissa
- precision
- point
- leading

Stack

Don't say:
- LIFO
- push
- pop
- top

De Morgan's

Don't say:
- bar
- NOT
- swap
- AND/OR

Hash table

Don't say:
- key
- function
- collision
- index

Linked list

Don't say:
- node
- pointer
- next
- traverse

Logical shift

Don't say:
- bits
- multiply
- divide
- zero

Binary search tree

Don't say:
- left
- right
- subtree
- ordered

Bingo — word bank & ready-to-cut cards

Word bank (students fill a blank 3×3 grid, or use the pre-filled cards below): binary 00001010, binary 00010100, binary 00011110, binary 00101000, binary 01100100, binary 11111111, binary 10000000, binary 01000000, binary 00010000, binary 00000001, hex 0F, hex 1E, hex 32, hex 64, hex FF, hex 80, hex 0A, hex 2D, hex C8, hex 19

BINGO 1

hex 64	binary 00101000	hex 32
hex 1E	binary 00011110	binary 01000000
hex 80	binary 00010000	binary 10000000

BINGO 2

binary 00010100	binary 10000000	binary 00011110
hex 32	binary 11111111	hex C8
hex 1E	hex 80	binary 00000001

BINGO 3

hex 0A	hex 64	binary 01000000
binary 01100100	hex 19	hex 80
hex 2D	binary 00011110	hex 0F

BINGO 4

hex 80	hex 2D	hex 1E
binary 01000000	hex FF	hex 0A
binary 00010000	hex 19	binary 10000000

BINGO 5

hex 64	binary 00101000	hex 2D
hex 0A	binary 10000000	hex 32
hex 19	hex FF	hex C8

BINGO 6

hex 80	hex 32	hex 2D
hex 64	binary 00000001	binary 11111111
binary 00010000	hex 0F	binary 00011110

Bingo — caller's list (teacher: read the clue, students cross off the term)

#	Clue to read aloud	Answer
1	Ten	binary 00001010
2	Twenty	binary 00010100
3	Thirty	binary 00011110
4	Forty	binary 00101000
5	One hundred	binary 01100100
6	The largest value in one byte (unsigned)	binary 11111111
7	Two to the power 7	binary 10000000
8	Two to the power 6	binary 01000000
9	Two to the power 4	binary 00010000
10	The smallest positive whole number	binary 00000001
11	Hex F in denary	hex 0F
12	Fifty	hex 32
13	Forty-five	hex 2D
14	Two hundred	hex C8
15	Twenty-five	hex 19

Quick-fire — clue & answer (self-test / quiz-quiz-trade)

#	Clue	Answer
1	The signed representation where the MSB has a negative place value.	Two's complement
2	Convert binary 11001010 to denary.	202
3	Convert denary 45 to hex.	2D
4	The float part that determines the range of representable values.	Exponent
5	A right shift that copies the sign bit in from the left.	Arithmetic (right) shift
6	LIFO data structure with push, pop and peek.	Stack
7	Convert hex BE to denary.	190
8	An ordered, immutable sequence such as (x, y).	Tuple
9	The half-adder output given by A AND B.	Carry
10	Two keys mapping to the same index in a hash table.	Collision
11	Convert binary 1111 to hex.	F
12	An in-order traversal of a BST visits the nodes in this order.	Ascending order